Bell's Inequality

and the EPR experiment

→ interpretation: local hidden variables

① 'bra-ket':

$$\langle v| = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

bra

$$|v\rangle = (v_1^* \; v_2^* \; v_3^* \; v_4^*)$$

ket

so: $\langle u|v\rangle =$ " dot product "

inner product

$$\langle u|v\rangle = |v_1|^2 + |v_2|^2 + |v_3|^2 + |v_4|^2$$

e.g. $\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\langle u|v\rangle = x^2 + y^2 + z^2$

imagine a continuous function $f(x)$ is like a vector w/ ∞ dimensions

$$|f\rangle = \begin{pmatrix} f(x_1) \\ f(x_2) \\ f(x_3) \\ \vdots \end{pmatrix}$$

$$\langle f| = (\ldots f^*(x_1)dx, f^*(x_2)dx, f^*(x_3)dx, \ldots)$$

$$p = -i\hbar\vec{\nabla}$$

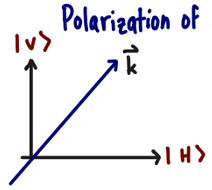
$$\frac{f(x+dx) - f(x)}{dx}, \ldots$$

where:

$$\langle f|f\rangle = \int |f(x)|^2 dx \; : \; f = \psi(x)$$

$$\langle f|f\rangle = \int |\psi(x)|^2 dx$$

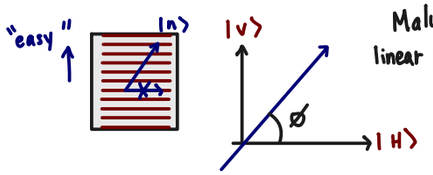
Polarization of E&M Waves



General polarization Vector:

$|n\rangle = \alpha|H\rangle + \beta|v\rangle$

↑ complex number



Malus' Law

linear polarizer: input field has angle θ relating to 'easy' polarization

$E_0 \cos\theta$

intensity: $\propto (E_0 \cos\theta)^2$

↳ $I_f = I_i \cos^2(\phi)$

• introduce polarization measurement operator

$$\hat{P} = \begin{pmatrix} \cos^2\phi & \sin\phi\cos\phi \\ \sin\phi\cos\phi & \sin^2\phi \end{pmatrix}$$

$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ e.g. $\vec{P_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \cos^2\phi \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

want: $\hat{P}_1|\psi\rangle = |\psi'\rangle$

$$\langle\psi'|\psi'\rangle = \frac{1}{2}(\cos^2\phi + \sin^2\phi)^2 = \frac{1}{2} \quad \left(\text{if } \theta_1 - \theta_2 = \frac{\pi}{2}\right)$$

$$\hat{P}_2\hat{P}_1|\psi\rangle = |\psi\rangle$$

$$\underbrace{\langle\psi''|\psi''\rangle = \frac{1}{2}\cos^2(\phi_1 - \phi_2)}_{\text{coincidence rate of each photon getting through polarizer}}$$

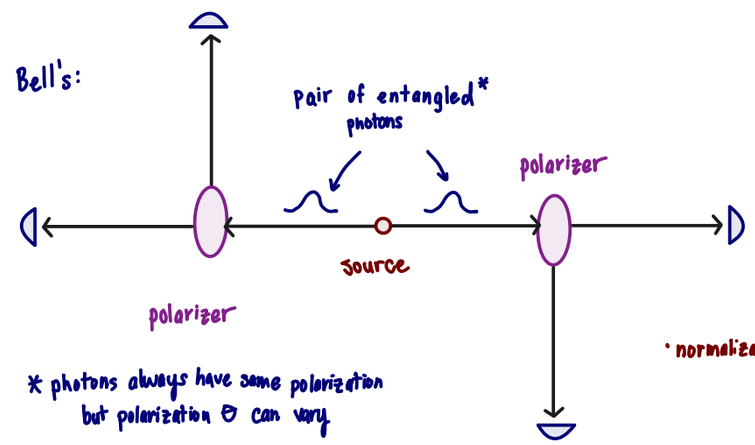
Source Generates Entangled Pairs

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|H\rangle|H\rangle + |v\rangle|v\rangle)$$

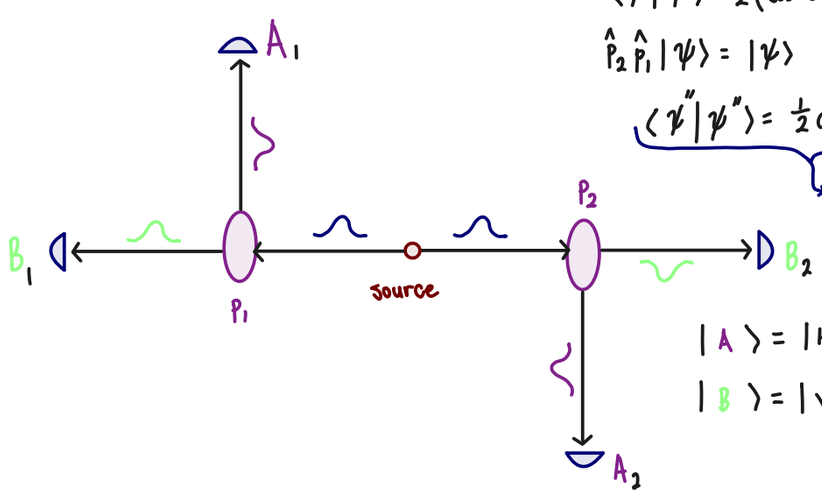
leftward particle

rightward particle

Bell's:



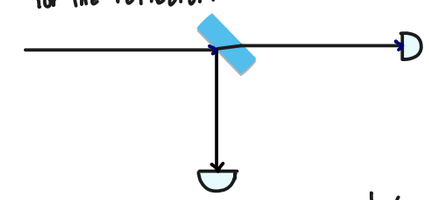
* photons always have same polarization but polarization θ can vary



$|A\rangle = |H\rangle$

$|B\rangle = |v\rangle$

for the reflector:



output from the beamsplitter: $|\psi\rangle = \frac{1}{\sqrt{2}}(|1\rangle|0\rangle + |0\rangle|1\rangle)$

left detector

right detector

• normalization: $\langle\psi|\psi\rangle = \frac{1}{\sqrt{2}}(\langle 1|\langle 0| + \langle 0|\langle 1|)\frac{1}{\sqrt{2}}(|1\rangle|0\rangle + |0\rangle|1\rangle)$

$$= \frac{1}{2}(\underbrace{\langle 1|1\rangle}_{=1}\underbrace{\langle 0|0\rangle}_{=0} + \langle 1|0\rangle\langle 0|1\rangle)$$

for the 'expected' case

'pure random'

↳ we would expect: either $|H\rangle|H\rangle$ or $|v\rangle|v\rangle$ } 50% of both

the reality is, due to entanglement, we resolve 0% of both